# RobinReal.ai

*Datathon 2026 - Analytics Club ETH Zurich*

## Neural Nap Team

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PROBLEM

# Traditional Search Fails to Capture Real Intent

### Rigid Filters

Keyword-based SQL queries miss semantic intent and soft preferences entirely.

### Unstructured Intent

Natural language queries carry location, budget, and lifestyle signals simultaneously.

### Our Goal

Bridge the gap between free-text queries and structured listing databases.

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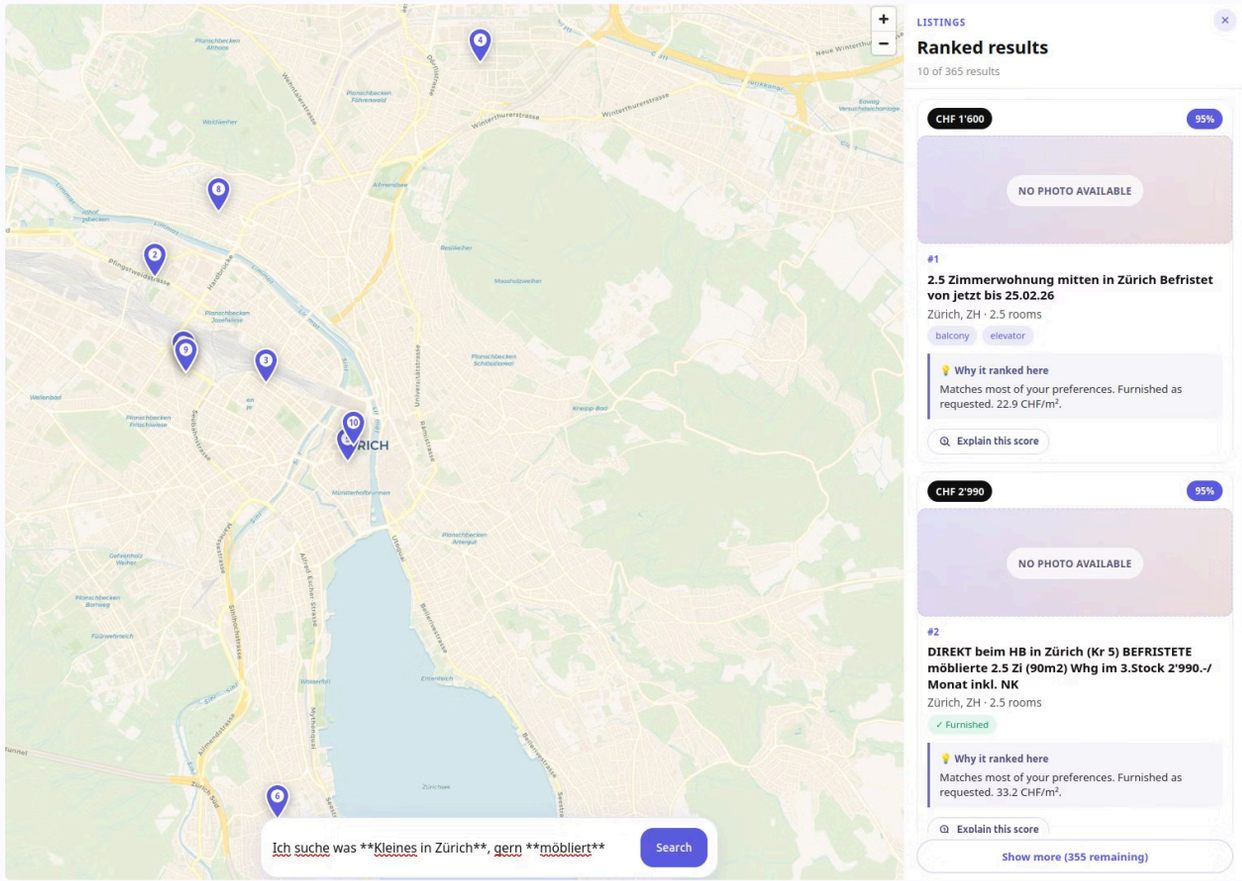
# Introducing RobinReal.ai: Your Conversational Real Estate Engine

*Experience your search results come to life, beautifully visualized on an interactive map.*

- Simply describe your ideal home
- Identification of both hard constraints and soft preferences.
- Intelligent listings and AI-generated explanations for each match.
- The whole engine is exposed as an MCP (Model Context Protocol) server — meaning it can be called natively by Claude and other AI hosts as a tool.

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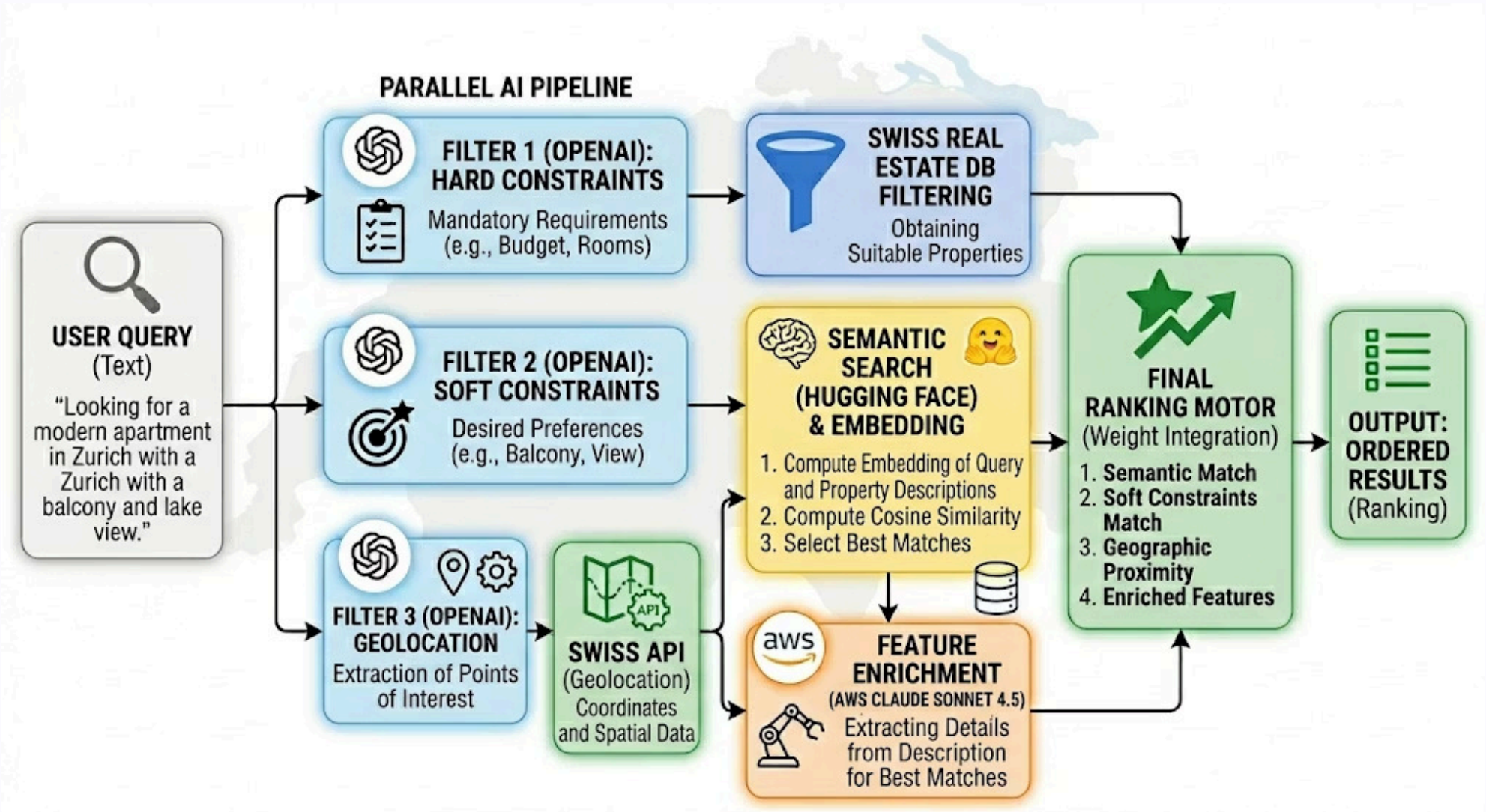
# Pipeline

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# Multi-Signal Ranking Engine

Our sophisticated ranking engine harmonizes three distinct signals to deliver highly relevant property recommendations:

### Semantic Similarity

Leveraging embeddings, we capture the core meaning and nuanced intent of user queries, moving beyond rigid keywords.

### Geospatial Proximity

Precise location scoring using ensures properties are within desired geographic bounds.

### Soft Preferences

An LLM is strategically deployed to extract subjective attributes like "bright" or "modern" that embeddings often miss.

The final score is a weighted sum of these signals. These weights are **query-adaptive**, dynamically reconfiguring based on what the user emphasizes. For instance, adding a location in a query significantly boosts the geospatial weight, ensuring location relevance. Every recommendation comes with **full transparency**, detailing exactly which features contributed to its score.

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# Results & Key Takeaways

**~6K**

### Swiss Listings

Indexed and searchable across German, French, and English queries.

**25+**

### LLM Features

Structured attributes extracted per listing description.

**3**

### Ranking Signals

Semantic, geolocation, and soft-constraint scores combined dynamically.

→ **Dynamic Weight Allocation**

Ranking adapts automatically to query complexity.

→ **LLM-Gated Extraction**

Cost-controlled, threshold-based invocation keeps latency low.

→ **Next Steps: Pre-compute description features offline**

Instead of calling the LLM at query time, extract all features from listing descriptions in advance and store them directly in the database. This removes the LLM from the hot path entirely, makes the database richer, and speeds up the ranking pipeline significantly.

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